Fourier Transforms

Let $F(x)$ be a function defined on $(-\infty, \infty)$ and piecewise continuous in each finite partial interval and integrable in $(-\infty, \infty)$ then;

$$f(p) = F\{F(x)\} = \int_{-\infty}^{\infty} e^{-ipx} . F(x)\, dx$$

is called the Fourier transform of $f(x)$, and is denoted by $f(p)$.

The function $F(x)$ is called the inverse Fourier transform of $f(p)$ and is written as;

$$F(x) = F^{-1}\{f(p)\}$$

$$F(x) = F^{-1}\{f(p)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ipx} f(p)\, dp.$$

Q. Show that the fourier transform $f(x) = e^{-x^2/2}$ is $\sqrt{2\pi}\, e^{-p^2/2}$.

We have;

$$F\{F(x)\} = \int_{-\infty}^{\infty} e^{-ipx} . e^{-x^2/2}\, dx$$

$\cancel{\int_0^{\infty} e^{-p^2/2} d}$

$$= \int_{-\infty}^{\infty} e^{-(x+ip)^2/2} . e^{-p^2/2}\, dx$$

$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$ Ans.

$$= e^{-p^2/2}\int_{-\infty}^{\infty} e^{-(x+ip)^2/2}\, dx$$

Putting $x + ip = \sqrt{2}y \Rightarrow dx = \sqrt{2}\, dy$

$$= e^{-p^2/2}\int_{-\infty}^{\infty} e^{-(\sqrt{2}y)^2/2}\sqrt{2}\, dy$$

$$= \sqrt{2}\, e^{-p^2/2}\int_{-\infty}^{\infty} e^{-y^2} dy$$

$$= \sqrt{2} \times e^{-p^2/2} \times 2 \times \int_0^{\infty} e^{-y^2} dy = \sqrt{2}\, e^{-p^2/2} . \cancel{2} . \frac{\sqrt{\pi}}{\cancel{2}}$$

$$= \sqrt{2\pi}\, e^{-p^2/2}$$

Ans.

Q. Find the Fourier transform of

$$F(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a \end{cases}$$

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$$\therefore \quad F\{F(x)\} = \int_{-\infty}^{\infty} e^{-ipx} \cdot F(x)\, dx$$

$$= \int_{-\infty}^{-a} e^{-ipx} \cdot 0 \cdot dx + \int_{-a}^{a} e^{-ipx} \cdot x\, dx + \int_{+a}^{\infty} e^{-ipx} \cdot 0 \cdot dx$$

$$F\{F(x)\} = \int_{-a}^{a} \underset{II}{e^{-ipx}} \cdot \underset{I}{x} \cdot dx$$

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$= \left[x \cdot \frac{e^{-ipx}}{(-ip)}\right]_{-a}^{a} + \frac{1}{ip}\int_{-a}^{a} e^{-ipx}\, dx$$

$$= \left[x \cdot \frac{e^{-ipx}}{-ip}\right]_{-a}^{a} + \left(+\frac{e^{-ipx}}{\cancel{i^2}\, p^2}\right)_{-a}^{a}$$

$$= \left[\frac{a \cdot e^{-ipa}}{-ip} + \left(\frac{+a \cdot e^{ipa}}{-ip}\right)\right] + \left(\frac{e^{-ipa}}{p^2} - \frac{e^{ipa}}{p^2}\right)$$

$$= e^{-ipa}\left[\frac{1}{p^2} - \frac{a}{ip}\right] - \frac{e^{ipa}}{p}\left[\frac{1}{p} + \frac{1}{i}\right]$$

$$= \frac{a \cdot e^{-ipa}}{p}\left[\frac{1}{p} - \frac{a}{i}\right] - \frac{e^{ipa}}{p}\left[\frac{1}{p} + \frac{1}{i}\right] \quad \text{Ans.}$$

OR

$$= \frac{a}{-ip}\left[e^{-ipa} + e^{ipa}\right] + \frac{1}{p^2}\left[e^{-ipa} - e^{+ipa}\right]$$

$$= \frac{2a}{-ip}\cos pa - \frac{1}{p^2}\, 2i\sin pa = \frac{2}{p}\left[\frac{a\cos pa}{-i} - \frac{i\sin pa}{p}\right] \quad \text{Ans.}$$

Q. Find the fourier transform of

$$F(x) = \begin{cases} 1-x^2, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$$

$$\therefore \quad F\{F(x)\} = \int_{-\infty}^{\infty} e^{-ipx} \cdot F(x)\, dx$$

$$= \int_{-1}^{1} e^{-ipx}(1-x^2)\, dx$$

$$= \int_{-1}^{1} e^{-ipx}\, dx - \int_{-1}^{1} e^{-ipx} \cdot x^2 \cdot dx$$

Let $I_1 = \int_{-1}^{1} \underset{II}{e^{-ipx}} \cdot \underset{I}{x^2}\, dx$

$$= \left[x^2 \cdot \frac{e^{-ipx}}{-ip}\right]_{-1}^{1} - \int_{-1}^{1} 2x \cdot \frac{e^{-ipx}}{-ip}\, dx$$

$\therefore$ Let $I_2 = \int_{-1}^{1} \underset{II}{e^{-ipx}} \cdot \underset{I}{x} \cdot dx$

$$I_2 = \left[x \cdot \frac{e^{-ipx}}{-ip}\right]_{-1}^{1} - \int_{-1}^{1} \frac{e^{-ipx}}{-ip}\, dx$$

$$= \left[x \cdot \frac{e^{-ipx}}{-ip}\right]_{-1}^{1} - \left[\frac{e^{-ipx}}{(ip)^2}\right]_{-1}^{1}$$

$$I_2 = \left[\frac{1 \cdot e^{-ip}}{-ip} - \left(\frac{-1 \cdot e^{ip}}{-ip}\right)\right] + \frac{1}{p^2}\left[e^{-ip} - e^{ip}\right]$$

$$\therefore I_1 = \left[\frac{e^{-ip}}{-ip} + \frac{e^{ip}}{ip}\right] + \frac{2}{ip}\left[\left(\frac{-e^{-ip}}{ip} - \frac{e^{ip}}{ip}\right) + \left(\frac{e^{-ip} - e^{ip}}{p^2}\right)\right]$$

$$I = \left(\frac{e^{-ipx}}{-ip}\right)_{-1}^{1} - I_1$$

$$= \left(\frac{e^{-ip}}{-ip} + \cancel{\frac{e^{ip}}{+ip}}\right) - \left(\frac{e^{-ip}}{-ip} + \cancel{\frac{e^{ip}}{ip}}\right) - \frac{2}{ip}\left[\left(\frac{-e^{-ip} - e^{ip}}{ip}\right) + \left(\frac{e^{-ip} - e^{ip}}{p^2}\right)\right]$$

$$= \frac{2}{ip^2}\left[\left(\frac{e^{-ip} + e^{ip}}{i}\right) - \left(\frac{e^{-ip} - e^{ip}}{p}\right)\right]$$

$$= \frac{2}{ip^2}\left[\frac{2\cos p}{i} + \frac{2i\sin p}{p}\right] \quad \text{Ans}$$

$$= -\frac{4\cos p}{p^2} + \frac{4\sin p}{p^3} \quad \text{Ans}$$

+ Fourier Sine Transform

The fourier-sine transform of a function $f(x)$, $0 < x < \infty$ denoted by $F_s\{F(x)\}$ or $f_s(p)$ is defined as;

$$F_s\{F(x)\} = f_s(p) = \int_0^\infty F(x) \sin px \, dx$$

The function $f(x)$ is called the inverse fourier sine transform of $f_s(p)$ and is written as;

$$F(x) = F_s^{-1}\{f_s(p)\}$$

It is given by the so-called inversion formula

$$F(x) = F_s^{-1}\{f_s(p)\} = \frac{2}{\pi}\int_0^\infty f_s(p) \sin px \, dp.$$

Q. Find the fourier transform of the

$$f(x) = \begin{cases} \sin x, & 0 < x < a \\ 0, & \text{when } x > a \end{cases}$$

We have $F_s\{F(x)\} = \int_0^\infty F(x) \sin px \, dx$

$$= \int_0^a \sin x \cdot \sin px \, dx + \int_a^\infty 0 \cdot \sin px \, dx$$

$$= \frac{1}{2} \int_0^a (\cos(x - px) - \cos(x + px)) \, dx$$

$$= 2 \left[\frac{\sin(1-p)x}{1-p} - \frac{\sin(1+p)x}{1+p} \right]_0^a$$

$$F_s\{F(x)\} = 2 \left[\frac{\sin(1-p)a}{1-p} - \frac{\sin(1+p)a}{1+p} \right]$$ Ans.

Fourier Cosine Transform:

The Fourier cosine transform of a function $F(x)$, $0 < x < \infty$ denoted by $F_s\{F(x)\}$ or $f_c(p)$ is defined as:

$$F_c\{F(x)\} = f_c(p) = \int_0^a F(x) \cos px \, dx$$

The function $f(x)$ is called the inverse fourier cosine transform of $f_c(p)$ and is written as;

$$F(x) = F_c^{-1}\{f_c(p)\}$$

It is given by the so-called inversion formula

$$F(x) = F_c^{-1}\{f_c(p)\} = \frac{2}{\pi} \int_0^\infty f_c(p) \cos px \, dp.$$

Q. Find the fourier cosine transform of

$$f(x) = \begin{cases} \cos x, & 0 < x < a \\ 0, & x > a \end{cases}$$

We have $F_c\{F(x)\} = \int_0^a F(x) \cos px \, dx$

$$= \int_0^a \cos x \cos px \, dx$$

$$= \frac{1}{2} \int [\cos(1-p)x + \cos(1+p)x] \, dx$$

$$= \frac{1}{2} \left[\frac{\sin(1-p)x}{(1-p)} + \frac{\sin(1+p)x}{(1+p)} \right]_0^a$$

$$F_c\{F(x)\} = \frac{1}{2} \left[\frac{\sin(1-p)a}{1-p} + \frac{\sin(1+p)a}{1+p} \right] \quad \text{Ans.}$$

Q. Find the Fourier Cosine transform of

$$f_c(p) = \int_0^\infty e^{-x^2} \cos px \, dx = I$$

Differentiating w.r.t p under the sign of integration

$$\frac{\partial I}{\partial p} = \int_0^\infty e^{-x^2} \frac{\partial \cos px}{\partial p} \cdot dx = \int_0^\infty e^{-x^2} (-x \sin px) \, dx$$

$$\frac{\partial I}{\partial p} = \frac{1}{2} \int_0^\infty 2 \left(e^{-x^2} (-x \sin px) \right) dx$$

$$= - \int_0^\infty \Big(\underbrace{(e^{-x^2} x)}_{\mathrm{II}} \underbrace{(\sin px)}_{\mathrm{I}} \Big) \, dx$$

$$\frac{\partial I}{\partial p} = - \left[\frac{1}{2} \left(-\sin px \, / \, e^{-x^2} \right)_0^\infty + p \int_0^\infty \frac{\cos px}{p} \cdot e^{-x^2} dx \right]$$

0

$$\Rightarrow \frac{\partial I}{\partial p} = -pI$$

$$\therefore \frac{\partial I}{I} = -p\,\partial p$$

Integrating both sides

$$\log I = \frac{-p^2}{2} + \log A$$

$$\log I - \log A = -p^2/2$$

$$\log \frac{I}{A} = -p^2/2$$

$$\frac{I}{A} = e^{-p^2/2}$$

$$\therefore I = A e^{-p^2/2} \quad \text{——} \quad (1)$$

When $p = 0$

$$I = \int_0^\infty e^{-x^2} \cos px \, dx$$

$$= \int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

For $p = 0 \Rightarrow I = \sqrt{\pi}/2$

$$\sqrt{\pi}/2 = A e^0$$

$$A = \sqrt{\pi}/2$$

$$\Rightarrow I = \sqrt{\pi}/2 \, e^{-p^2/2}$$ Ans.

Linearity Property

If c_1 and c_2 are arbitrary constant, then ;

(i) $F\{c_1 f_1(x) + c_2 f_2(x)\} = c_1 F\{F_1(x)\} + c_2 F\{F_2(x)\}$

(ii) $F_s\{c_1 f_1(x) + c_2 F_2(x)\} = c_1 F_s\{F_1(x)\} + c_2 F_s\{F_2(x)\}$

(iii) $F_c\{c_1 F_1(x) + c_2 F_2(x)\} = c_1 F_c\{F_1(x)\} + c_2 F_s\{F_2(x)\}$

Q. Find the sine and cosine transform of

$$2e^{-5x} + 5e^{-2x}.$$

$\Rightarrow F_s\{2e^{-5x} + 5e^{-2x}\} = 2F_s\{e^{-5x}\} + 5F_s\{e^{-2x}\}$

Formula : $\int_0^\infty e^{-ax} \cdot \sin bx\,dx = \frac{b}{a^2+b^2}$

$$\int_0^\infty e^{-ax} \cdot \cos bx\,dx = \frac{a}{a^2+b^2}$$

$\Rightarrow F_s\{e^{-5x}\} = \int_0^\infty e^{-5x} \cdot \sin px\,dx = \frac{p}{p^2+25}$

$F_s\{e^{-2x}\} = \int_0^\infty e^{-2x} \cdot \sin px\,dx = \frac{p}{p^2+4}$

$F_s\{2e^{-5x} + 5e^{-2x}\} = \frac{2p}{p^2+25} + \frac{5p}{p^2+4}$ Ans.

and

$F_c\{2e^{-5x} + 5e^{-2x}\} = \frac{2\times(-5)}{p^2+25} + \frac{5\times(-2)}{p^2+4}$

$$= \frac{-10}{p^2+25} - \frac{10}{p^2+4}$$ Ans.

Change of scale property

If $f(p)$ is the fourier transform of $f(x)$, then $\frac{1}{a} f\left(\frac{p}{a}\right)$ is the fourier transform of $F(ax)$.

Proof: We have

$$F\{F(ax)\} = \int_{-\infty}^{\infty} e^{-ipx} \cdot F(ax)\,dx$$

Putting $ax = t$

$x = t/a \Rightarrow dx = \frac{dt}{a}$

$$\Rightarrow F\{F(ax)\} = \frac{1}{a}\int_{-\infty}^{\infty} e^{-ip\frac{t}{a}} \cdot F(t)\,dt$$

$$F\{F(ax)\} = \frac{1}{a} f\left(\frac{p}{a}\right) \quad \text{Proved.}$$

=> Shifting Property and Modulation Theorem

If $f(p)$ is the Fourier transform of $F(x)$ then $e^{-iap} f(p)$ is the Fourier transform of $F(x-a)$

$$F\{F(x-a)\} = e^{-iap} f(p)$$

$$\therefore \quad F\{F(x-a)\} = \int_{-\infty}^{\infty} e^{-ipx} \cdot F(x-a)\,dx$$

Let $x - a = t \Rightarrow x = t + a$

$dx = dt$

$$= \int_{-\infty}^{\infty} e^{-ipt} \cdot e^{-ipa} \cdot F(t)\,dt$$

$$F\{F(x-a)\} = e^{-ipa} f(p) \quad \text{Proved.}$$

This is known as Modulation theorem.

$\Rightarrow$ If $f(p)$ is the Fourier transform of $F(x)$, then show that

$$F\{F(x)\cos ax\} = \frac{1}{2}\{f(p-a)+f(p+a)\}$$

$$F\{F(x)\cos ax\} = \int_{-\infty}^{\infty} e^{-ipx} \cdot F(x) \cdot \cos ax \, dx$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} e^{-ipx} \cdot F(x) \cdot (e^{iax} + e^{-iax})\, dx$$

$$= \frac{1}{2}\left[\int_{-\infty}^{\infty} e^{-ix(p-a)} F(x)\, dx + \int_{-\infty}^{\infty} e^{-ix(p+a)} F(x)\, dx\right]$$

$$= \frac{1}{2}\{f(p-a)+f(p+a)\} \quad \text{Proved.}$$

$$\cos ax = \frac{e^{iax}+e^{-iax}}{2}.$$

+ Fourier transform of Derivative

If $f(p)$ is the Fourier transform of $F(x)$, then prove that if $f(p)$ is the Fourier transform of $F'(x)$.

By definition;

$$F\{F'(x)\} = \int_{-\infty}^{\infty} e^{-ipx} \cdot F'(x)\, dx$$

$$= \int_{-\infty}^{\infty} e^{-ipx}\left[\lim_{h\to 0} \frac{F(x+h)-F(x)}{h}\right] dx$$

$$= \cancel{\int_{-\infty}^{\infty} \frac{1}{h}} \lim_{h\to 0}\left[\frac{1}{h}\left\{\int_{-\infty}^{\infty} e^{-ipx} \cdot F(x+h)\, dx - \int_{-\infty}^{\infty} e^{-ipx} F(x)\, dx\right\}\right]$$

Let $x+h = t$

$x = t-h$

$dx = dt$

$$= \lim_{h \to 0} \left\{ \frac{1}{h} \left[\int_{-\infty}^{\infty} e^{-ip(t-h)} F(t)\,dt - f(p) \right] \right\}$$

$$= \lim_{h \to 0} \left\{ \frac{1}{h} \left[e^{iph} f(p) - f(p) \right] \right\}$$

$$\therefore = \lim_{h \to 0} \frac{f(p)[e^{iph} - 1]}{h} \quad : \frac{0}{0} \text{ form}$$

Using L'Hospital Rule

$$\Rightarrow \lim_{h \to 0} \frac{ip\, f(p)\, e^{iph}}{1} = ip\, f(p)$$

$$\Rightarrow F\{F'(x)\} = ip\, f(p)$$ Proved.

⇒ If $f(p)$ is the Fourier transform of $F(x)$

$$\Rightarrow F\{F^{n}(x)\} = (ip)^{n} f(p).$$

⇒ Convolution Theorem

If $F\{f(x)\}$ and $F\{g(x)\}$ are fourier transform of the functions $f(x)$ and $g(x)$ respectively, then Fourier transform of the convolution of $f(x)$ and $g(x)$ is the product of thier fourier transform.

$$F\{f(x) * g(x)\} = F\{f(x)\} \cdot F\{g(x)\}$$

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Q. Find Fourier sine transform of
$f(x) = 1/x$.

$$F_s\{f(x)\} = \int_0^\infty f(x).\ \sin px\, dx$$

$$= \int_0^\infty \frac{\sin px}{x}\, dx = \pi/2 \quad \text{Ans.}$$

~~As (p > 0)~~ As $(p > 0)$

Since,

$$\int_0^\infty \frac{\sin mx}{m} = \begin{cases} \pi/2, \text{ if } m > 0 \\ -\pi/2, \text{ if } m < 0 \end{cases}$$

=> Finite Fourier transforms and Fourier Integrals

⇒ Finite Fourier Sine Transform:
If $f(x)$ is piecewise continuous function in the interval $(0, l)$ then the finite Fourier sine transform is denoted by $F_s\{F(x)\}$ or $f_s(p)$ and is defined as

$$F_s\{F(x)\} = f_s(p) = \int_0^l F(x) \sin\left(\frac{p\pi x}{l}\right) dx, \text{ where } p \text{ is an integer.}$$

And inverse formula;

$$F(x) = F^{-1}\{f_s(p)\} = \frac{2}{l} \sum_{p=1}^\infty f_s(p) \sin\left(\frac{p\pi x}{l}\right).$$

⇒ Finite Fourier Cosine Transform:
If $f(x)$ is piecewise continuous function in the interval $(0, l)$ then the finite Fourier cosine transform is denoted and defined as:

$$F_c\{F(x)\} = f_c(p) = \int_0^l F(x).\ \cos\left(\frac{p\pi x}{l}\right) dx, \text{ where } p \text{ is an integer}$$

Inverse formula;

$$F(x) = F^{-1}\{f_c(p)\} = \frac{2}{l} \sum_{p=1}^\infty f_c(p) \cos\left(\frac{p\pi x}{l}\right).$$

Q. Find the finite Fourier sine and cosine transforms of $f(x) = 1$, $(0, \pi)$.

$$\therefore \quad f_s(p) = \int_0^\pi 1 \cdot \sin\left(\frac{p\pi x}{\pi}\right)dx = \left[-\frac{\cos(px)}{p}\right]_0^\pi$$

$$= -\frac{\cos(p\pi)}{p} + \frac{1}{p} = \frac{-(-1)^p + 1}{p} \quad \text{Ans.}$$

$$f_c(p) = \int_0^\pi 1 \cdot \cos\left(\frac{p\pi x}{\pi}\right)dx = \left[\frac{\sin(px)}{p}\right]_0^\pi = 0 \quad \text{Ans.}$$

=> Fourier Integrals

⇒ Fourier Integral Theorem:

If $f(x)$ is a function Dirichlet's condition in every finite interval $(-l, +l)$ and $\int^\infty |F(x)|dx$ converges i.e $f(x)$ is absolutely integrable in the interval $(-\infty, +\infty)$ then

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t)\cos u(t-x)\,du\,dt$$ i.e ↳ Fourier Integral

$$f(x) = \frac{1}{2\pi} \times 2 \times \int_0^\infty\int_{-\infty}^{\infty} f(t)\cos u(t-x)\,du\,dt$$

⇒ Fourier Sine integral formula

The fourier sine integral formula is given by

$$f(x) = \frac{2}{\pi}\int_0^\infty\int_0^\infty f(t)\sin ut \sin ux\,du\,dt$$

→ Fourier Cosine Integral Formula

The fourier cosine integral formula is given by

$$f(x) = \frac{2}{\pi}\int_0^\infty\int_0^\infty f(t)\cos ut \cos ux \, du\, dt$$

→ Fourier complex integral formula

The fourier complex integral formula is given by

$$f(x) = \frac{2}{\pi}\int_{-\infty}^\infty\int_{-\infty}^\infty f(t)\, e^{iut} \cdot e^{-iux}\, du\, dt$$

The integral on the right hand side is called the exponential form of the Fourier Integral.

⇒ Application of Fourier Transform to Boundary Value Problem

$$F_s\left\{\frac{\partial^2 u}{\partial x^2}\right\} = p(u)_{x=0} - p^2\bar{u}_s \quad \text{where } \bar{u}_s = F_s(u)$$

$$F_c\left\{\frac{\partial^2 u}{\partial x^2}\right\} = -\left(\frac{\partial u}{\partial x}\right) - p^2\bar{u}_c \quad \text{where } \bar{u}_c = F_c(u)$$

Q. Solve $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $x > 0$, $t > 0$ subject to conditions

$u(0,t) = 0$, $u(x,0) = \begin{cases} 1, & 0 < x < 1 \\ 0, & x > 1 \end{cases}$, $u(x,t)$ is bounded.

$$\therefore \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

Applying Fourier sine transform

$$\Rightarrow F_s\left\{\frac{\partial u}{\partial t}\right\} = F_s\left\{\frac{\partial^2 u}{\partial x^2}\right\}$$

$$\int_0^\infty \frac{\partial u}{\partial t}\sin px\, dx = \cancel{u(0,t)} - p^2 \bar{u}_s$$

$\quad 0$

$$\frac{\partial}{\partial t}\int_0^\infty u \sin px \, dx = -p^2 \bar{u}_s$$

$$\frac{\partial}{\partial t}\bar{u}_s = -p^2 \bar{u}_s$$

$$\frac{\partial \bar{u}_s}{\bar{u}_s} = -p^2 \partial t$$

Integrating both sides

$$\log \bar{u}_s = -p^2 t + \log A$$

$$\log \bar{u}_s - \log A = -p^2 t$$

$$\log \frac{\bar{u}_s}{A} = -p^2 t$$

$$\bar{u}_s = A e^{-p^2 t}$$

$\therefore$ If $t = 0$

$$\Rightarrow \bar{u}_s(x, 0) = A e^{-p^2(0)}$$

$$A = \int_0^\infty u_{(x,0)} \sin px \, dx$$

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$$\cancel{A = \int_0^\infty u}$$

As $u = 1$; $0 < x < 1$

and $u = 0$, $x > 0$

$$\Rightarrow \quad A = \int_0^1 1 \cdot \sin px \, dx + \int_1^\infty \underbrace{0 \cdot \sin px \, dx}_{0}$$

$$A = \left[-\frac{\cos px}{p} \right]_0^1$$

$$A = \frac{1 - \cos px}{p}$$

$$\Rightarrow \quad \bar{U}_s = \left(\frac{1 - \cos px}{p} \right) e^{-p^2 t}$$

$$F_s\{u\} = \left(\frac{1 - \cos px}{p} \right) e^{-p^2 t}$$

Applying inverse sine Fourier transform

$$u = F_s^{-1} \left\{ \left(\frac{1 - \cos px}{p} \right) e^{-p^2 t} \right\}$$ Ans.

$$u = \frac{2}{\pi} \int_0^\infty \left(\frac{1 - \cos px}{p} \right) e^{-p^2 t} \sin px \, dp$$

Ans.

Q. Solve

$$\frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2} \quad \text{for} \quad 0 \leq x < \infty, \ t > 0$$

Given the conditions;

(i) $u(x, 0) = 0$ for $x \geq 0$

(ii) $\frac{\partial u}{\partial x}(0, t) = -\mu$ (constant)

(iii) $u(x, t)$ is bounded.

Sol. Since,

$u'(0, t)$ i.e $(U')_{x=0}$ is given

Taking Fourier cosine transform both sides

$$\therefore \quad F_s\left\{\frac{\partial u}{\partial t}\right\} = F_s\left\{K\frac{\partial^2 u}{\partial x^2}\right\}$$

$$\Rightarrow \int_0^\infty \frac{\partial u}{\partial t} \sin px \, dx = K \int_0^\infty \frac{\partial^2 u}{\partial x^2} \sin px \, dx$$

$$\frac{\partial}{\partial t}\int_0^\infty u \sin px \, dx = K\left\{\left(-\frac{\partial u}{\partial x}\right)_{x=0} - p^2 \bar{u}_c\right\}$$

$$\therefore \quad \frac{\partial \bar{u}_c}{\partial t} = K\left\{+\mu - p^2 \bar{u}_c\right\}$$

$$\Rightarrow \frac{\partial \bar{u}_c}{\partial t} + Kp^2 \bar{u}_c = K\mu,$$ we get a linear differential equation.

$$\frac{\partial \bar{u}_c}{\partial t} + P\bar{u}_c = Q$$

Calculating.

$$I.F = e^{\int P\,dt} = e^{\int K p^2 dt} = e^{Kp^2 t}.$$

$\therefore$ The solution is written as;

$$\bar{u}_c \times IF = \int Q \times I.F \times dt$$

$$\bar{u}_c \times e^{Kp^2 t} = \int K\mu \times e^{Kp^2 t}\,dt$$

$$\bar{u}_c \times \cancel{e^{Kp^2 t}} = \frac{\cancel{K}\mu \times \cancel{e^{Kp^2 t}}}{\cancel{K}p^2} + C$$

$$\therefore \quad \bar{\mu}_c = \frac{\mu}{p^2} + C e^{-Kp^2 t} \quad \text{—— (1)}$$

Putting $t=0$ in equation (1)

$$\therefore \quad \bar{\mu}_c(p,0) = \frac{\mu}{p^2} + C$$

$$\int_0^\infty \underset{(t=0)}{\cancel{\mu}} \cos px\,dx = \frac{\mu}{p^2} + C$$

0

$$\Rightarrow \quad \boxed{C = \frac{-\mu}{p^2}}$$

Putting value of C in (1), we get

$$\Rightarrow \quad \bar{\mu}_c = \frac{\mu}{p^2} - \frac{\mu}{p^2} e^{-Kp^2 t}$$

Taking Inverse Fourier cosine transform both sides.

$$F_c^{-1}\{\bar{\mu}_c\} = F_c^{-1}\left\{\frac{\mu}{p^2}\left[1 - e^{-Kp^2 t}\right]\right\}$$

$$\mu = \frac{2}{\pi}\int_0^\infty \frac{\mu}{p^2}\left[1 - e^{-Kp^2 t}\right]\cos px\,dp \quad \underline{\underline{\text{Ans}}}.$$

Q. Solve

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad x > 0,\ t > 0 \text{ subject to}$$

conditions;

$$u_x(0,t) = 0$$

$u(x,t)$ is bounded and

$$u(x,0) = \begin{cases} x, & 0 \leq x \leq 1 \\ 0, & x > 1 \end{cases}$$

As, $u_x(0,t)$ i.e $\frac{\partial u}{\partial x}(0,t)$ is given.

Taking Fourier Cosine transform both sides, we get

$$F_c\left\{\frac{\partial u}{\partial t}\right\} = F_c\left\{\frac{\partial^2 u}{\partial x^2}\right\}$$

$$\therefore \int_0^\infty \frac{\partial u}{\partial t} \cos px\, dx = \int_0^\infty \frac{\partial^2 u}{\partial x^2} \cos px\, dx$$

Now, $$\frac{\partial}{\partial t}\int_0^\infty u \cos px\, dx = -\underbrace{\left(\frac{\partial u}{\partial x}\right)_{x=0}}_{0} - p^2 \bar{u}_c$$

$$\frac{\partial \bar{u}_c}{\partial t} = -p^2 \bar{u}_c$$

$$\therefore \frac{\partial \bar{u}_c}{\bar{u}_c} = -p^2 dt$$

Integrating both sides, we get

$$\Rightarrow \quad \log \bar{u}_c = -p^2 t + \log C$$

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$$\log \bar{u}_c - \log C = -p^2 t$$

$$\log \frac{\bar{u}_c}{C} = -p^2 t$$

$$\bar{u}_c = C e^{-p^2 t} \quad \text{——— (1)}$$

Putting $t=0$ in equation (1), we get

$$\bar{u}_c(p,0) = C$$

$$\therefore \quad C = \int_0^\infty u(x,0) \cos px \, dx$$

$$C = \int_0^1 \underset{I}{x} \underset{II}{\cos px} \, dx + \int_1^\infty 0 \cdot \cos px \, dx$$

$$C = \left[x \cdot \frac{\sin px}{p} \right]_0^1 - \int_0^1 1 \cdot \frac{\sin px \, dx}{p}$$

$$= \left[\frac{x \cdot \sin px}{p} \right]_0^1 - \left[\frac{-\cos px}{p^2} \right]_0^1$$

$$C = \frac{\sin p}{p} + \left[\frac{\cos p}{p^2} - \frac{1}{p^2} \right]$$

Putting the value of C in eq (1)

$$\therefore \quad \bar{u}_c = \left[\frac{\sin p}{p} + \frac{1}{p^2}(\cos p - 1) \right] e^{-p^2 t}$$

Taking Inverse Fourier cosine transform, we get

$$u = \frac{2}{\pi} \int_0^\infty \left[\frac{\sin p}{p} + \frac{1}{p^2}(\cos p - 1) \right] e^{-p^2 t} \cos px \, dp \quad \text{Ans.}$$

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Q. Using suitable fourier transform. Solve

$$\frac{\partial u}{\partial t} = 2\frac{\partial^2 u}{\partial x^2} \quad \text{if}$$

$$u(0,t) = 0,$$
$$u(x,0) = e^{-x},$$
$$u(x,t) \text{ is bounded.}$$

As, $u_{x=0}$ i.e $u(0,t)$ is given.

Taking Fourier Sine transform both sides

$$F_s\left\{\frac{\partial u}{\partial t}\right\} = F_s\left\{2 \times \frac{\partial^2 u}{\partial x^2}\right\}$$

$$\therefore \int_0^\infty \frac{\partial u}{\partial t}\sin px\,dx = 2\int_0^\infty \frac{\partial^2 u}{\partial x^2}\sin px\,dx$$

$$\frac{\partial}{\partial t}\int_0^\infty u\sin px\,dx = 2\left\{p\underbrace{u_{x=0}}_{0} - p^2\bar{u}_s\right\}$$

$$\frac{\partial \bar{u}_s}{\partial t} = 2 \times p^2 \times \bar{u}_s$$

$$\therefore \frac{\partial \bar{u}_s}{\bar{u}_s} = 2p^2\partial t$$

Integrating both sides, we get

$$\log \bar{u}_s = 2p^2 t + \log C$$
$$\log \bar{u}_s - \log C = 2p^2 t$$
$$\log \frac{\bar{u}_s}{C} = 2p^2 t$$

$$\bar{U}_s = C e^{-2p^2 t} \quad \text{———} \quad (1)$$

Putting $t = 0$ in equation (1)

$$\therefore \quad \bar{U}_s(p, 0) = C$$

$$\int_0^\infty u(x,0) \sin px \, dx = C$$

$$\therefore \quad \int_0^\infty e^{-x} \sin px \, dx = C$$

Let $\quad I = \displaystyle\int_0^\infty \underset{II}{e^{-x}} \underset{I}{\sin px} \, dx$

$$\Rightarrow \quad I = \left[\frac{\sin px \, e^{-x}}{-1} \right]_0^\infty + \int_0^\infty \overset{I}{p \cos px} \, \overset{II}{\frac{e^{-x}}{+1}} \, dx$$

$$I = 0 + p \left[\left[-\cos px \, e^{-x} \right]_0^\infty + p \int_0^\infty -\sin px \, e^{-x} \, dx \right]$$

$$I = p\left[-0 + (1)\right] + p^2(-I)$$

$$\therefore \quad I = \frac{p}{1+p^2}$$

$$\Rightarrow \quad \boxed{C = \frac{p}{1+p^2}}$$

Putting C in eq (1)

$$\therefore \quad \bar{U}_s = \left(\frac{p}{1+p^2} \right) e^{-2p^2 t}.$$

Taking Inverse Fourier Sine transforms.

$$u = \frac{2}{\pi} \int_0^\infty \frac{p}{1+p^2} \cdot e^{-2p^2 t} \sin px \, dp \quad \text{Ans.}$$

⇒ Application of Finite Fourier transform to B.V.P

The finite fourier transform can be applied to the boundary value problems in which the range of one variable is finite.

To determine finite fourier transform of partial derivative.

(i) Consider the first order partial derivative $\frac{\partial u}{\partial x}$ where u is a function of x and t and $0 < x < l$ and $t > 0$, we have

$$F_s\left\{\frac{\partial u}{\partial x}\right\} = \int_0^l \frac{\partial u}{\partial x} \sin\frac{p\pi x}{l}\, dx = -\frac{p\pi}{l} F_c(u)$$

$$F_c\left\{\frac{\partial u}{\partial x}\right\} = \int_0^l \frac{\partial u}{\partial x} \cos\frac{p\pi x}{l}\, dx = \frac{p\pi}{l} F_s(u) - \left\{u(0,t) - (-1)^p u(l,t)\right\}$$

(ii) Consider the second order order partial derivative $\frac{\partial^2 u}{\partial x^2}$ when u is a function of x and t and $0 < x < l$ and $t > 0$, we have

$$F_s\left\{\frac{\partial^2 u}{\partial x^2}\right\} = -\frac{p\pi}{l}\left[\frac{p\pi}{l} F_s(u) - \left\{u(0,t) - (-1)^p u(l,t)\right\}\right]$$

$$F_c\left\{\frac{\partial^2 u}{\partial x^2}\right\} = \frac{p\pi}{l} F_s\left(\frac{\partial u}{\partial x}\right) - \left\{u_x(0,t) - (-1)^p u_x(l,t)\right\}$$

Note: Choice of finite sine or cosine transform

The choice of finite sine or cosine transformed depend upon the form of boundary conditions

1. Finite sine transform: If $u(0,t)$ and $u(l,t)$ are known.
2. Finite cosine transform: If $u_x(0,t)$ and $u_x(l,t)$ are known.

Q. Solve

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 6, \ t > 0$$

$$u(0,t) = 0, \quad u(6,t) = 0 \quad \text{and} \quad u(x,0) = \begin{cases} 1, & 0 < x < 3 \\ 0, & 3 < x < 6 \end{cases}$$

As, $u(0,t)$ and $u(l,t)$ are known.

Taking finite Fourier sine transform on both sides, we get

$$F_s\left\{\frac{\partial u}{\partial t}\right\} = F_s\left\{\frac{\partial^2 u}{\partial x^2}\right\}$$

$$\therefore \int_0^6 \frac{\partial u}{\partial t} \sin\frac{p\pi x}{6}\, dx = \int_0^6 \frac{\partial^2 u}{\partial x^2} \sin\frac{p\pi x}{6}\, dx$$

$$\frac{\partial}{\partial t}\int_0^6 u \sin\frac{p\pi x}{6}\, dx = -\frac{p\pi}{6}\left[\frac{p\pi}{6} F_s(u) - \{\underbrace{u(0,t)}_{0} - (-1)^p \underbrace{u(6,t)}_{0}\}\right]$$

$$\frac{\partial \bar{u}_s}{\partial t} = -\frac{p\pi}{6}\left[\frac{p\pi}{6}\bar{u}_s - 0\right]$$

$$\therefore \frac{\partial \bar{u}_s}{\partial t} = -\left(\frac{p\pi}{6}\right)^2 \bar{u}_s$$

$$\frac{\partial \bar{u}_s}{\bar{u}_s} = -\left(\frac{p\pi}{6}\right)^2 \partial t$$

Integrating both sides, we get

$$\Rightarrow \log \bar{u}_s = -\left(\frac{p\pi}{6}\right)^2 t + \log C$$

$$\log \bar{u}_s - \log C = \left(-\left(\frac{p\pi}{6}\right)^2\right)t$$

$$\log \frac{\bar{u}_s}{C} = -\left(\frac{p\pi}{6}\right)^2 t$$

$$\bar{u}_s = C e^{-\left(\frac{p\pi}{6}\right)^2 t} \quad \text{——— } (1)$$

Putting $t=0$ in equation (1)

$$\therefore \bar{u}_{s(p,0)} = C$$

$$\therefore C = \int_0^6 u_{(x,0)} \sin \frac{p\pi x}{6}\, dx$$

$$C = \int_0^3 u(x,0) \sin \frac{p\pi x}{6}\, dx + \int_3^6 u(x,0) \sin \frac{p\pi x}{6}\, dx$$

$$C = \int_0^3 \sin \frac{p\pi x}{6}\, dx$$

$$C = \left[\frac{-\cos \frac{p\pi x}{6}}{\frac{p\pi x}{6}}\right]_0^3 = \frac{6}{-p\pi}\left[\cos\left(\frac{p\pi}{2}\right) - 1\right]$$

Putting C in equation (1)

$$\bar{u}_s = \left\{\frac{-6}{p\pi}\left[\cos\left(\frac{p\pi}{2}\right) - 1\right]\right\} e^{\left(-\frac{p\pi}{6}\right)^2 t}$$

Now, taking Inverse finite fourier sine transform, we get

$$u = \frac{1}{3}\sum_{p=1}^{\infty}\left[\frac{-6}{p\pi}\left[\cos\left(\frac{p\pi}{2}\right)-1\right]e^{-(p\pi/6)^2 t}\right]\sin\left(\frac{p\pi x}{6}\right) \quad \text{Ans.}$$

- Fourier Series

- Periodic Functions

A function $f(x)$ is said to be periodic with period P, if for each x in the domain of $f(x)$, the point $x+P$ is also in the domain of $f(x)$ such that

$$f(x+P) = f(x)$$

The smallest positive number P if it exists with this property is called the Fundamental Period.

The sine and cosine function are periodic with period 2π.

$$\sin(x+2\pi) = \sin x.$$

- Even and Odd functions

A function $f(x)$ is said to be even if $f(-x) = f(x)\ \forall\ x$ in the domain of $f(x)$.

Similarly,
$f(x)$ is said to be odd if $f(-x) = -f(x)\ \forall\ x$ in the domain of $f(x)$.

Theorem: If a periodic function $f(x)$ with period 2π is the sum function of an uniformly convergent series given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos nx + b_n\sin nx\right] \quad (1)$$

in the closed interval $-\pi \leqslant x \leqslant \pi$ then

$$\left.\begin{aligned} a_0 &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx \\ a_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \\ b_n &= \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \end{aligned}\right\} \quad (2)$$

Q. Find the fourier series for $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$.

Deduce that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots = \frac{\pi^2}{8}$

$$\therefore \quad f(x) = \frac{a_0}{2} + \sum_{n=1}^{n=\infty} [a_n \cos nx + b_n \sin nx]$$

$$\Rightarrow \quad a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{0} (-\pi)\, dx + \int_{0}^{\pi} x\, dx \right]$$

$$= \frac{1}{\pi} \left[(-\pi) (x)_{-\pi}^{0} + \left[\frac{x^2}{2}\right]_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[(-\pi)[0 - (-\pi)] + \left[\frac{\pi^2}{2} - 0\right] \right]$$

$$= \frac{1}{\pi} \left[-\pi^2 + \frac{\pi^2}{2} \right]$$

$$\therefore \quad a_0 = \frac{-\pi}{2}$$

Now,

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{0} (-\pi) \cos nx\, dx + \int_{0}^{\pi} \underset{I}{x} \underset{II}{\cos nx}\, dx \right]$$

$$= \frac{1}{\pi} \left[(-\pi) \left[\frac{\sin nx}{n}\right]_{-\pi}^{0} + \left[\frac{x \sin nx}{n}\right]_0^{\pi} - \int_0^{\pi} \frac{\sin nx}{n}\, dx \right]$$

$$= \frac{1}{\pi} \left[(-\pi)(0-0) + \left[\frac{\pi(0)}{n} - 0\right] + \left[\frac{\cos nx}{n^2}\right]_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\frac{\cos n\pi}{n^2} - \frac{1}{n^2} \right] = \frac{1}{\pi} \left[\frac{(-1)^n - 1}{n^2} \right].$$

and $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$

$$b_n = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-\pi)\sin nx\,dx + \int_{0}^{\pi}\underset{I}{x}\,\underset{II}{\sin nx}\,dx\right]$$

$$= \frac{1}{\pi}\left[(-\pi)\left[-\frac{\cos nx}{n}\right]_{-\pi}^{0} + \left[\frac{x(-\cos nx)}{n}\right]_{0}^{\pi} + \int_{0}^{\pi}\frac{\cos nx}{n}dx\right]$$

$$= \frac{1}{\pi}\left[\pi\left[\frac{1}{n} - \frac{(-1)^n}{n}\right] + \left[-\frac{\pi(-1)^n}{n}\right] + \left[\frac{\sin nx}{n^2}\right]_{0}^{\pi}\right]$$

$$= \frac{1}{\cancel{\pi}}\left[\frac{\cancel{\pi}}{n} - \frac{\cancel{\pi}(-1)^n}{n} - \frac{\cancel{\pi}(-1)^n}{n} + 0\right]$$

$$b_n = \frac{1-2(-1)^n}{n}$$

$$\therefore f(x) = \frac{-\pi}{4} + \sum_{n=1}^{\infty}\frac{((-1)^n-1)}{\pi n^2}\cos nx + \sum_{n=1}^{\infty}\frac{1-2(-1)^n}{n}\sin nx$$

$$\therefore f(x) = \frac{-\pi}{4} + \left[\frac{-2}{\pi}\frac{\cos 1x}{1^2} + \frac{-2}{\pi}\frac{\cos 3x}{3^2} + \dots\right] + \left[\frac{3}{1}\sin 1x + \frac{-1}{2}\sin 2x + \dots\right]$$

Putting $x = 0$, we get

$$f(0) = \frac{-\pi}{4} - \frac{2}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right] + [0+0+\dots]$$

Now, as $f(x)$ is discontinuous at $x = 0$,

Calculating limits;

$$f(0^-) = \lim_{h\to 0} f(0-h) = -\pi$$

$$f(0^+) = \lim_{h\to 0} f(0+h) = 0$$

$\therefore$ The sum of the series at $x = 0$ is

$$f(0) = \frac{1}{2}\left[f(0^-) + f(0^+)\right]$$

$$= \frac{1}{2}\left[-\pi + 0\right] = -\frac{\pi}{2}$$

$$\Rightarrow \quad -\frac{\pi}{2} = -\frac{\pi}{4} - \frac{2}{\pi}\left[\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right]$$

$$\therefore \quad \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$$ Ans.

$\Rightarrow$ Fourier series for even function

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where
$$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$$

and
$$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx$$

$\Rightarrow$ Fourier series for odd function

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where
$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\,dx$$

Q. Find the fourier series of the function $f(x)$ with period 2π such that $f(x) = x^2$, $-\pi \leq x \leq \pi$.
Hence, find the sum of the series

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots$$

Here, $f(x)$ is an even function so it's fourier series is cosine series of the form:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

$\therefore$ $$a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx = \frac{2}{\pi}\int_0^{\pi} x^2 dx = \frac{2}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{2\pi^2}{3}.$$ Ans.

$$a_n = \frac{2}{\pi}\int_0^{\pi} \underset{I}{x^2}\,\underset{II}{\cos nx}\,dx$$

$$= \frac{2}{\pi}\left[\left[x^2 \frac{\sin nx}{n}\right]_0^{\pi} - \int_0^{\pi} \frac{\overset{I}{2x}\,\overset{II}{\sin nx}}{n}dx\right]$$

$$= \frac{2}{\pi}\left[0 - \frac{2}{n}\left[\left[x.\left(-\frac{\cos nx}{n}\right)\right]_0^{\pi} + \int_0^{\pi}\frac{\cos nx}{n}\right]\right]$$

$$= \frac{2}{\pi}\left[-\frac{2}{n}\left[\frac{-\pi(-1)^n}{n} + \left[\frac{\sin nx}{n}\right]_0^{\pi}\right]\right]$$

$$a_n = \frac{2}{\cancel{\pi}}\left[\frac{2\cancel{\pi}(-1)^n}{n^2}\right] = \frac{4(-1)^n}{n^2}$$

$$\therefore\ f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty}\frac{4(-1)^n}{n^2}\cos nx$$

$$f(x) = \frac{\pi^2}{3} + 4\left[-\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \dots\right]$$

Putting $x = 0$

$$0 = \frac{\pi^2}{3} - 4\left[\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots\right]$$

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$ Ans.

Q. Find the fourier series corresponding to the function $f(x)$.

$$f(x) = \begin{cases} 2, & \text{if } -2 < x \leq 0 \\ x, & \text{if } 0 < x < 2 \end{cases}$$

Let the fourier series of $f(x)$ be of the formula ;

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\frac{n\pi x}{l} + b_n \sin\frac{n\pi x}{l}\right] \quad \text{—— (1)}$$

Here , $l = 2$.

$$\Rightarrow \quad a_0 = \frac{1}{l}\int_{-l}^{l} f(x)\,dx = \frac{1}{2}\int_{-2}^{2} f(x)\,dx$$

$$= \frac{1}{2}\int_{-2}^{0} 2\,dx + \frac{1}{2}\int_{0}^{2} x\,dx$$

$$= [0 - (-2)] + \frac{2^2}{2\cdot 2}$$

$$a_0 = 2 + 1 = 3 \quad \text{Ans.}$$

and

$$a_n = \frac{1}{l}\int_{-l}^{l} f(x)\cos\frac{n\pi x}{l}\,dx$$

$$a_n = \frac{1}{2}\int_{-2}^{0} f(x)\cos\frac{n\pi x}{2}\,dx + \frac{1}{2}\int_{0}^{2} f(x)\cos\frac{n\pi x}{2}\,dx$$

$$= \frac{1}{2}\int_{-2}^{0} 2\cos\frac{n\pi x}{2}\,dx + \frac{1}{2}\int_{0}^{2} \underset{I}{x}\,\overset{II}{\cos\frac{n\pi x}{2}}\,dx$$

$$= \left[\frac{2\sin\frac{n\pi x}{2}}{n\pi}\right]_{-2}^{0} + \frac{1}{2}\left[\left[\frac{2x}{n\pi}\sin\frac{n\pi x}{2}\right]_{0}^{2} + \left[\left(\frac{2}{n\pi}\right)^2\cos\left(\frac{n\pi x}{2}\right)\right]_{0}^{2}\right]$$

$$= 0 + 0 + \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right] \quad \text{Ans.}$$

$$\therefore \quad b_n = \frac{1}{l}\int_{-l}^{l} f(x)\sin\frac{n\pi x}{l}\,dx$$

$$b_n = \frac{1}{2}\int_{-2}^{0} 2\sin\frac{n\pi x}{2}\,dx + \frac{1}{2}\int_{0}^{2} \overset{\mathrm{I}}{x}\,\overset{\mathrm{II}}{\sin}\frac{n\pi x}{2}\,dx$$

$$= \left[\frac{-\cos\left(\frac{n\pi x}{2}\right)}{\left(\frac{n\pi}{2}\right)}\right]_{-2}^{0} + \frac{1}{2}\left[\left[\frac{2x}{n\pi}\left(-\cos\frac{n\pi x}{2}\right)\right]_0^2 + \left[\left(\frac{2}{n\pi}\right)^2\sin\frac{n\pi x}{2}\right]_0^2\right]$$

$$= -\frac{2}{n\pi}\left[1-(-1)^n\right] + \frac{1}{2}\left[-\frac{4}{n\pi}(-1)^n + 0\right]$$

$$= -\frac{2}{n\pi} + \cancel{\frac{2}{n\pi}(-1)^n} - \cancel{\frac{2}{n\pi}(-1)^n}$$

$$b_n = -\frac{2}{n\pi}$$

$$\Rightarrow f(x) = \frac{3}{2} + \sum_{n=1}^{\infty}\left[\frac{2}{n^2\pi^2}\left[(-1)^n-1\right]\cos\frac{n\pi x}{2} + \left(\frac{-2}{n\pi}\right)\sin\frac{n\pi x}{2}\right] \quad \text{Ans.}$$

$\Rightarrow$ Half Ranged Expansion

Suppose a function $f(x)$ is defined in the interval $0 \le x \le l$, then defining this function in the arbitrary manner on the interval $-l \le x \le 0$ we can expand it in a fourier series we consider the following two particular case.

Case 1: Half Range Cosine series in $0 \le x \le l$

If the function is defined on the interval by $f(-x) = f(x)$ then we shall obtain an even function.

This is called the even extension.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos\frac{n\pi x}{l}, \quad 0 \le x < l$$

$$a_0 = \frac{2}{l}\int_0^l f(x)\,dx \quad \text{and}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx, \quad n = 1, 2, \ldots$$

The series is called Half-ranged cosine series in $[0, l]$.

Case 2: Half ranged sine series in the interval $0 \leq x \leq l$.

If the function $f(x)$ is defined in the interval $-l \leq x \leq 0$ by $f(x) = -f(x)$ then we shall obtain an odd function.

This is called the odd extension.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}, \quad 0 \leq x \leq l$$

where

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

Q. Expand the function $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2-x, & 1 < x < 2 \end{cases}$ in the interval $(-2, 2)$ in a series of sine and also in a series of cosine.

Hence, we know series of cosine is given as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l}$$

$$\therefore \quad a_0 = \frac{2}{l}\int_0^l f(x)\,dx = \frac{2}{2}\left[\int_0^1 x\,dx + \int_1^2 (2-x)\,dx\right]$$

$$= \left[\frac{x^2}{2}\right]_0^1 + \left[2x - \frac{x^2}{2}\right]_1^2$$

$$= \frac{1}{2} + \left[(4-2) - (2-1/2)\right] = \frac{1}{2} + \cancel{2} - \cancel{2} + \frac{1}{2} = 1 \text{ Ans.}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\,dx$$

$$\therefore a_n = \int_0^1 x\cos\frac{n\pi x}{2}\,dx + \int_1^2 (2-x)\cos\frac{n\pi x}{2}\,dx$$

$$a_n = \left[\frac{2x}{n\pi}\sin\frac{n\pi x}{2}\right]_0^1 + \left[\left(\frac{2}{n\pi}\right)^2\cos\frac{n\pi x}{2}\right]_0^1 + \left[\frac{2\times 2}{n\pi}\sin\frac{n\pi x}{2}\right]_1^2$$

$$- \left[\left[\frac{2x}{n\pi}\sin\frac{n\pi x}{2}\right]_1^2 + \left[\left(\frac{2}{n\pi}\right)^2\cos\frac{n\pi x}{2}\right]_1^2\right]$$

$$a_n = \frac{2}{n\pi}\cancel{\sin\frac{n\pi}{2}} + \left[\left(\frac{2}{n\pi}\right)^2\left[\cos\frac{n\pi}{2} - 1\right]\right] + \left[\frac{4}{n\pi}(0) - \cancel{\frac{4}{n\pi}\sin\frac{n\pi}{2}}\right]$$

$$- \left[\left[\frac{4}{n\pi}(0) - \cancel{\frac{2}{n\pi}\sin\frac{n\pi}{2}}\right] + \left[\left[\frac{2}{n\pi}\right]^2\left[(-1)^n - \cos\frac{n\pi}{2}\right]\right]\right]$$

$$a_n = \left(\frac{2}{n\pi}\right)^2\left[2\cos\frac{n\pi}{2} - (-1)^n - 1\right]$$ Ans.

$\therefore$ The cosine series is written as :

$$f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{4}{n^2\pi^2}\left[2\cos\frac{n\pi}{2} - (-1)^n - 1\right]\cos\frac{n\pi x}{2}$$ Ans.

$$\cancel{f(x) = \frac{1}{2} + \left[\frac{4}{1^2\pi^2}\left[2(0) - (-1) \cdots\right] + \frac{4}{2^2\pi^2}\left[2(-1) - \cdots\right] + \cdots\right]}$$

~~Ans.~~

Now, the sine series is given as;

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l}$$

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx$$

$$\therefore b_n = \frac{2}{2}\int_0^2 f(x)\sin\frac{n\pi x}{2}\,dx$$

$$b_n = \int_0^1 \overset{I}{x} \, \overset{II}{\sin} \frac{n\pi x}{2} \, dx + \int_1^2 \overset{I}{(2-x)} \, \overset{II}{\sin} \frac{n\pi x}{2} \, dx$$

$$= \left[\frac{2x}{n\pi} \left(-\cos \frac{n\pi x}{2} \right) \right]_0^1 + \left[\left(\frac{2}{n\pi} \right)^2 \sin \frac{n\pi x}{2} \right]_0^1 + \left[\frac{2(2-x)}{n\pi} \left(-\cos \frac{n\pi x}{2} \right) \right]_1^2$$

$$- \left[\left(\frac{2}{n\pi} \right)^2 \sin \frac{n\pi x}{2} \right]_1^2$$

$$= -\frac{2}{n\pi} \cancel{\cos \frac{n\pi}{2}} + \left(\frac{2}{n\pi} \right)^2 \sin \frac{n\pi}{2} - 2(0) + \frac{2}{n\pi} \cancel{\cos \frac{n\pi}{2}} - 0 + \left(\frac{2}{n\pi} \right)^2 \sin$$

$$b_n = \frac{8}{n^2 \pi^2} \sin \frac{n\pi}{2}$$ Ans.

$$\therefore \quad f(x) = \sum_{n=1}^{\infty} \frac{8}{n^2 \pi^2} \sin \frac{n\pi}{2} \sin \frac{n\pi}{2} x$$ Ans.

Parseval's Identity for Fourier Transforms

Theorem: If $f(p)$ and $g(p)$ are complex fourier transform of $F(x)$ and $G(x)$ respectively, then

(i) $$\frac{1}{2\pi}\int_{-\infty}^{\infty} f(p)\,\overline{g(p)}\,dp = \int_{-\infty}^{\infty} F(x)\,\overline{G(x)}\,dx$$

(ii) $$\frac{1}{2\pi}\int_{-\infty}^{\infty} |f(p)|^2\,dp = \int_{-\infty}^{\infty} |F(x)|^2\,dx$$

Here bar represents complex conjugate.

Some more results:

(i) $$\frac{2}{\pi}\int_{0}^{\infty} f_c(p)\,\overline{g_c(p)}\,dp = \int_{0}^{\infty} F(x)\,\overline{G(x)}\,dx$$

$$\frac{2}{\pi}\int_{0}^{\infty} f_s(p)\,\overline{g_s(p)}\,dp = \int_{0}^{\infty} F(x)\,\overline{G(x)}\,dx$$

(ii) $$\frac{2}{\pi}\int_{0}^{\infty} |f_c(p)|^2\,dp = \int_{0}^{\infty} |F(x)|^2\,dx$$

$$\frac{2}{\pi}\int_{0}^{\infty} |f_s(p)|^2\,dp = \int_{0}^{\infty} |F(x)|^2\,dx$$

Q. Using Parseval's identity to prove:

$$\int_{0}^{\infty} \frac{dt}{(a^2+t^2)(b^2+t^2)} = \frac{\pi}{2ab(a+b)}$$

Let $$\int_{0}^{\infty} \frac{dt}{(a^2+t^2)(b^2+t^2)} = \frac{1}{ab}\int_{0}^{\infty}\left(\frac{a}{a^2+t^2}\right)\left(\frac{b}{b^2+t^2}\right)dt$$

Let

$$F(x) = e^{-ax}$$

$$G(x) = e^{-bx}$$

$$\therefore F_c\{F(x)\} = F_c\{e^{-ax}\} = \int_0^\infty e^{-ax} \cos tx \, dx = \frac{a}{a^2+t^2}$$

$$F_c\{G(x)\} = F_c\{e^{-bx}\} = \int_0^\infty e^{-bx} \cos tx \, dx = \frac{b}{b^2+t^2}$$

$$\therefore \left[\frac{2}{\pi}\int_0^\infty f_c(t)\, g_c(t)\, dt\right] = \int_0^\infty F(x)\, \overline{G(x)}\, dx$$

$$= \int_0^\infty e^{-ax} \cdot e^{-bx}\, dx$$

$$= \int_0^\infty e^{-(a+b)x}\, dx$$

$$= \left[\frac{-e^{-(a+b)x}}{(a+b)}\right]_0^\infty$$

$$= \frac{1}{a+b}$$

$$\therefore \int_0^\infty \frac{dt}{(a^2+t^2)(b^2+t^2)} = \frac{\pi}{2ab}\left[\frac{2}{\pi}\int_0^\infty f_c(t)\, g_c(t)\, dt\right] = \frac{\pi}{2ab(a+b)}$$

Ans.

Q. Prove

$$\int_0^\infty \frac{x^2}{(x^2+1^2)^2}\,dx = \frac{\pi}{4}.$$

Sol.

$$\int_0^\infty \frac{x^2}{(x^2+1^2)^2} = \int_0^\infty \left(\frac{x}{x^2+1^2}\right)\left(\frac{x}{x^2+1^2}\right) dx$$

Let

$F(t) = e^{-t}$ and $G(t) = e^{-t}$, then

$$F_s\{F(t)\} = F_s\{e^{-t}\} = \frac{x}{x^2+1^2}$$

$$F_s\{G(t)\} = F_s\{e^{-t}\} = \frac{x}{x^2+1^2}$$

$$\therefore \frac{2}{\pi}\int_0^\infty f_s(t)\,\overline{g_s(t)}\,dt = \int_0^\infty F(t)\,\overline{G(t)}\,dx$$

$$= \int_0^\infty e^{-t}\cdot e^{-t}\,dt$$

$$= \int_0^\infty e^{-2t}\,dt$$

$$\frac{2}{\pi}\int_0^\infty \frac{x^2}{(1+x^2)^2}\,dx = \left[-\frac{e^{-2t}}{2}\right]_0^\infty$$

$$\int_0^\infty \frac{x^2}{(1+x^2)^2}\,dx = \frac{\pi}{4} \quad \text{Ans.}$$